

DRY-RUN GUIDE | 07_suffix_count.cpp
Source: cpp\2nd_belt\07_suffix_count.cpp

CORE BEHAVIOR

Reads words and a suffix, checks whether each word ends with that suffix using compare on the trailing substring, and counts matches.

REPRESENTATIVE INPUT / CALL

Input: N=4; words 'cat hat cats that'; suffix 'at'.

STEP TRACE

'cat', 'hat', and 'that' end in 'at'; 'cats' does not. Count=3.

EXPECTED OUTPUT / RETURN

'3'

ACTUAL SOURCE EXCERPT

Verbatim lines from the source; long lines wrap. Source files were not modified.

```
001 | /*
002 | Question:
003 | You are given a list of words and a suffix string. Your task is to count
004 | how many words in the list end with the given suffix.
005 |
006 | A suffix is a group of characters present at the end of a word.
007 |
008 | Print the total number of words that have the given suffix.
009 |
010 | Input:
011 | The first line contains an integer N - the number of words.
012 | The next input contains N words followed by the suffix string.
013 | */
014 |
015 | #include <iostream>
016 | #include <vector>
017 | #include <string>
018 | using namespace std;
019 |
020 | int main() {
021 |     int n;
022 |     cin >> n;
023 |
024 |     vector<string> words(n);
025 |     for (int i = 0; i < n; i++) {
026 |         cin >> words[i];
027 |     }
028 |
029 |     string suffix;
030 |     cin >> suffix;
031 |
032 |     int count = 0;
033 |
034 |     for (const string& word : words) {
035 |         if (word.size() >= suffix.size() &&
036 |             word.compare(word.size() - suffix.size(), suffix.size(), suffix) == 0) {
037 |             count++;
```

```
038 |         }
039 |     }
040 |
041 |     cout << count;
042 |
043 |     return 0;
044 | }
045 |
```